

Algorithmic Culture: Beyond the Photo/Film Divide

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In recent film and media studies, there has been a growing uneasiness among scholars with regard to the destiny of the moving image in the digital age. This article will argue that an investigation into the altered relationship between still and moving images can tell us a lot about the new condition of the moving image culture. Traditionally, the relationship between still and moving images has been governed by a discourse on the dialectical or antithetical relationship between film and photography. Historically speaking, the art of photography predates the invention of cinematography, but projectionists and filmmakers have always utilized this relationship dramatically and aesthetically as a transitional relationship between forms of mobility and immobility. Tom Gunning has shown how the Lumière brothers would astonish their audiences with a special technique of exhibition where they transform a projected still image into a moving image by cranking the projector. Since then, the ambiguity between stillness and motion has always been a resource for resonance and wonder in cinema and, in particular, in experimental film from the constructivists and surrealists all the way up to contemporary filmmakers like Jean-Luc Godard and Chris Marker in Europe and Ken Jacobs and Ernie Gehr in the United States. In more recent digital cinema (especially science fiction films), and in moving image installations in art galleries and on the Internet, a new kind of exploration of this resource is emerging. It may be connected to the transition from analog to digital.

Within the analog world of film stock and celluloid, the relationship between the still and the moving image was often looked upon as a way of foregrounding the material basis of the filmstrip, the relationship between the tacit frames and the act of projection. Accordingly, within the digital world of bits and bytes, the relationship between the still and the moving image has changed. But is this focus on stillness and motion outdated? On the contrary, it seems as if the relationship between stillness and motion has returned within a new and perhaps more complex matrix of problems. Several studies among acclaimed film scholars such as Laura Mulvey, Mary Ann Doane, Bernard Stiegler, Timothy Murray, Geoffrey Batchen, Thomas Elsaesser and W.J.T. Mitchell, have in recent years tried to diagnose the situation by reflecting on and doing close readings of certain works. Significantly, their studies are not simply close readings in the

formalist sense of the term; rather, they are investigations of a new situation. What has happened to the image? Scholars talk about a “naming crisis” in the realm of the digital image.¹ Is it still an image? Both scholars and artists seem to be urgently engaged in trying to assess this crisis – or new potential – of the image. A growing number of scholars such as Friedrich Kittler, Lev Manovich, D.N. Rodowick and Mark B.N. Hansen have also started to address these issues via the technical term “algorithm”, which was a term only used by computer specialists and mathematicians just a few years ago. This chapter will specifically address the relationship between new media, algorithms and the construction of the image. While many scholars trained in the world of analog and photochemical photography lament the introduction of digital image production because they think it impoverishes the experience of time and history, this paper will explore some of the new affordances of the digital in relation to aesthetic experience and sensation.

Even film scholars not interested in this aspect of the digital image see many advantages of digital technologies. Today, it is hard to imagine that before the advent of the laser disk and the videotape, most film scholars were writing their analyses from the memory of the film they had seen in a movie theatre. For instance, Stanley Cavell writes in the foreword to his seminal work, *The World Viewed* (1971), that, “I wrote primarily out of the memory of film”.² Laura Mulvey talks about a similar situation in the “Preface” to her *Death 24x a Second: Stillness and the Moving Image* (2006). These researchers and critics from the 1970s were confined to watching films in darkened rooms, projected at 24 (or thereabouts) frames per second. But new media changed this situation. “By the end of the twentieth century, ways of consuming cinema had multiplied and the regulation of its speed had been widely extended,” Mulvey writes.³ With the VCR, films could be stopped at any point, and with the DVD-player perfect freeze-frames were easily created. The pause button and the menu button have introduced new modes of analyzing film, thus opening up old films in new ways. In the essay, “Delaying cinema”, Mulvey contends:

The process of repetition and return involves stretching out the cinematic image to allow space and time for associative thought, reflection of resonance and connotation, the identification of visual clues, the interpretation of cinematic form and style, and, ultimately, personal reverie. Furthermore, by slowing down, freezing or repeating images, key moments and meanings become visible that could not have been perceived when hidden under the narrative flow and the movement of film. Although the alert spectator of melodrama may well have had the ability to read the cinematic language of displacement, consciously or subliminally, at 24 frames a second, today’s electronic or digital spectator can find these deferred meanings that have been waiting through the decades to be seen.⁴

Mulvey's narrative almost exhibits a teleology at play. She discusses the transition from the constrictions of the movie theatre spectator to the new revelatory potential of the "digital spectator" as critic. Through a series of brilliant readings, she shows how new technology can alter cinematic hermeneutics. Nevertheless, Mulvey's newly liberated observer addresses the "same" object, namely classical cinema. Her interests lie in the way new media can enhance and deepen the encounter with the unique singularity of the classic feature film. She is not interested in the potential of the new; she is interested in the potential of the old – finally revealed. She uses new media to study the potential of old media, without paying attention to how new media have changed the conditions of visual culture. Indeed, these two aspects of new media are interconnected. New media do not simply deal with or offer "the new". They address and, some would even say, strengthen the distribution and hegemony of the old. Many have already done encyclopedic assessments of what is new and what has remained the same since the digital revolution with regards to narrative cinema.⁵ What interests me here, is the changing vocation of the image after it became a technological object of a different kind, that is, a programmable object.

Programmable Objects

According to media philosopher Lev Manovich, a new media object is defined by its programmability.⁶ All programmable objects are composed of digital codes and are subject to algorithmic manipulation. The algorithm is the necessary conversion point in the transformation of the analog image into a digital image. Algorithms are abstract, symbolized, step-by-step instructions normally written in pseudo-code or drawn in flow-chart diagrams. Alan Turing, one of the inventors of the computer, described algorithms as any set of instructions fed into a machine to solve a problem.⁷ Put simply, algorithms describe how a computer can carry out the task you want it to do. They are expressed in a form called a program or software. They accept an input and produce an output. A text or an image that undergoes this kind of treatment, becomes a programmable object, or "softwareized".⁸ Within the visual field, the question of representation and indexicality is threatened. Now, is the new image "real" or not? Some hardcore celluloid fetishists, like D.N. Rodowick, would even venture to say that "digital depictions are not 'images', at least in the ordinary sense of the term".⁹ Even if digital imagining strives in most cases to be perceptually indistinguishable from the previous medium, namely, the photographic or photographic realism, the difference in the medium is crucial, and Rodowick's expla-

nation is worth listening to even if I disagree with some of his more pessimistic and nostalgic conclusions.

In *The Virtual Life of Film*, Rodowick discusses the differences between the analog and the digital as two distinct ontologies. While the analog photochemical process is based on a principle of continuity between input and output, the information processing of the digital image is, ontologically speaking, based on a separation or discontinuity between input and output. Without this fundamental discontinuity, computer algorithms wouldn't work. "The ontology of information processing...is agnostic with respect to its output," as Rodowick phrases it.¹⁰ It is the discontinuity between the input and the output that produces a new space for creativity and imagination – and chance and control. This space is addressed, not by light and shadow as in the photochemical process, but by computer algorithms, which carry out manipulations and alterations.

Algorithms is a key term here. Indeed, William Uricchio talks about today's "algorithmic turn" due to the new affordances of software such as photosynth. The "modern" *static* perspective, perfected by artists like Canaletto, isn't really challenged before the arrival of the "postmodern" *dynamic* and processual perspectives made possible by photosynth software, according to Uricchio.¹¹ But the algorithmic turn should also imply several elements of the less spectacular and the less visible. Alexander Galloway describes more generally *and* more specifically contemporary culture as an "algorithmic culture".¹² He uses the term to describe the culture of computer gaming as paradigmatic for understanding our new digital age. Video games are "informatic software" and belong to what he calls "algorithmic cultural objects".¹³ Gaming is "oriented around understanding and executing specific algorithms".¹⁴ In computer gaming, it is all about action and change. Each gesture, each action the player performs is connected to an algorithm. To Galloway, computer gaming becomes a metacritical reflection that reaches into the heart of computerized society.

Both Uricchio and Galloway make "processuality" and processing, that is, editing and otherwise manipulating existing digital images, central. Computer gaming, as an active intervention in the processual aspect of the technological image, becomes exemplary for the way more and more people interact, via algorithms, with available objects from our visual culture. Most of us do not write algorithms, we access them through application software. As I see it, playing around with digital editing software also is a "game" which is all about finding the best algorithm. Among the many well-known software packages are Adobe Photoshop and Final Cut Pro, which, in reality, consist of thousands of ready-made algorithms, and the user can activate these algorithms simply by importing an image and adding different layers or techniques using a variety of easily available tools to transform the image. The software industry even describes how their programs offer unique and robust graphics editing *experiences*. Some

of the most advanced effects deal with manipulating the quality of the “medium” itself. Indeed, what we see on the screen – whether it is a digital image that looks like a painting (after the “fresco” effect is applied), a sculpture, a mosaic, a still photograph or an image which is somehow moving or is a combination of still and moving elements (easy with the programming language called Processing) – is now just an option inscribed by the algorithms. In other words, the photo/film divide, which was closely associated within the analog culture with certain stable substances or medium-specific qualities such as celluloid, paper or chemistry, has become radically problematized. Indeed, the *same* digital information can be instantiated visually, acoustically, or kinesthetically with the help of adequate programming algorithms; the information can be warped, streamed, or sampled, accelerated or slowed down, supersaturated or attenuated. Programmable objects can simulate any previous media object. We can, in other words, call the algorithmic culture a *transversal* culture, because it so easily creates lines of flight between or across different “old” media. Certain medium-specific qualities have become preliminary appearances easily available for alterations.

Algorithmic cultures address a new materialism that is not based in medium-specific divisions between photo and film or still and moving images, for example, but rather in mutable information codes. Within a coded environment, forms of appearance are optional and governed by the new machinic agency of the viewer/user. This is the new situation established by the computer and reinforced further by the Web’s 2.0 culture, smarter software and more bandwidth. Film and photography are no longer medium-specific qualities, but are rather two of the ways that algorithms hide themselves. Significantly, we interact with the algorithms, but usually through an interface where the actual algorithmic instructions are hidden. As Galloway reminds us, now more than ever, the question is: *is what you see what you get?*¹⁵ The answer to this fundamental question is both yes and no. The coexistence of both of these phenomenologically incommensurable positions addresses both the problem and the potential of the new algorithmic culture.

The New Ground of the Image

The German artist Andreas Müller-Pohle has explored the relationship between image and information in its many variants. A crucial narrative behind his largely conceptual image production is the transition from the analog to the digital. In one of his most famous series of works, *Digital Scores (after Nicephore Niepce)* from 1995, he has literally turned visual culture upside down by folding

the computer language out of the black box (the machine) and into the white cube, the art gallery. In this piece, he reprocesses the information from the oldest existing photograph, Nicephore Niepce's so-called heliograph *View from the Window at Le Gras* from 1827. Niepce's heliography was made using a pewter plate coated with a solution of Bitumen of Judea, which was exposed in a camera for more than eight hours, then washed with oil of lavender and white petroleum to remove those parts of the bitumen not hardened by the light. To the left in the image, we see Niepce's pigeon house and to the right, a wing of the main building of his home. The light in this picture is magical. It is as if all of the normal shadows caused by the sun have been somehow erased. This is due to the fact that during the eight hours of exposure, the sun had crossed the sky and lit every object evenly. There is indeed *time* in this photograph, a time which encompasses a duration longer than most feature films.

In Andreas Müller-Pohle's *Digital Scores (after Nicephore Niepce)*, we no longer see Niepce's home or the pigeon house. The dots, the silver, the emulsion and the light are also gone. The visual information in the image has been turned into data, numerical representations. Müller-Pohle has taken the old image, the old medium of the heliograph, on a journey literally into the darkest corners of the digital machine. Niepce's heliograph (or at least the watercolor reproduction of it found in Helmut Gernsheim's 1983 *Geschichte der Photographie*) becomes a long winding digital code. Müller-Pohle displays the code on eight panels as a messy swarm of numbers and computational notations. Each panel represents one-eighth of one byte of memory. Niepce's eight hours of exposure has been transformed into eight panels of information. The time of the image has been spatialized into numeric code. In a gallery, the time of the image is "recaptured" as the time it takes to walk from one panel to the next.

Crucially, Müller-Pohle has reversed the image-making process and chosen to display the new matrix of the image. What kind of transformation are we witnessing? "We see not a photograph, but the new numerical rhetoric of photography", Geoffrey Batchen writes in a comment to this work.¹⁶ However, this numerical rhetoric should not be confused with the old relational geometry used in the rhetoric of composition, which goes back to the time of Alberti. We are instead dealing with an attempt to display the language of the computer as the new language of the image. The visual aesthetic of the original photograph has been decisively displaced in favor of the machine's capacity to "remember" the image as numbers to be manipulated. The result fiercely demonstrates the fundamental anti-aesthetic hidden in the logic of the algorithmic culture. Contemporary image processing usually involves digital conversion and attunement at some stage in its process. If this process is arrested to reveal the codes conditioning these operations, as in Müller-Pohle's case, the impression will denigrate the image as we know it. Some would even say that an algorithmic cul-

ture involves a fundamental critique of the visual layer of our culture. If visual culture is reborn as programmable codes, it is, at the same time, non-visual. In a questionnaire on the future of visual culture studies, Laura U. Marks asserted, "I personally think that the period of visual culture is over. Thus it's ironic that all these programs in visual studies are starting up, just at a point when information culture, which is invisible, is becoming the dominant form of our culture."¹⁷ Batchen thinks that information processing pushes us primarily into a kind of immaterialization:

At least since 1989 and the introduction of Adobe Photoshop into the marketplace, photography has been in the words of the French philosopher Jean-François Lyotard, an *immatériel* ("the principle on which the operational structure is based is not that of a stable 'substance', but that of an unstable ensemble of interactions ... the model of language"). Certainly, the identity of the photographic image no longer has to do with its support or its chemical composition, or with its authorship, place of origin, or pictorial appearance. It instead comprises, as Müller-Pohle suggests, a pliable sequence of digital data and electronic impulses. It is their configuration that now decides an image's look and significance, even the possibility of its continued existence. In other words, "photography" today is all about the reproduction and consumption, flow and exchange, maintenance and disruption, of data.¹⁸

I both agree and disagree with Batchen's radical conclusion. Since the initial stages of the digital revolution, we have learned that the immaterialization thesis of Lyotard and Batchen has its limitations. To a certain extent, it is due to a notion of programmability understood as conditioned by immateriality. It would be better to see the contemporary image as a technological object, which is processual. Even if digital photography comprises "a pliable sequence of digital data and electronic impulses", it still relies on a variable system of supports (capturing apparatuses with specific algorithms built in, software and hardware), which comprise the work process, or what Karl Marx would call "the hidden abode of production" where access is restricted or governed by what Friedrich Kittler has called "computeranalphabetismus".¹⁹ Rather than Batchen and Lyotard's focus on the process of immaterialization with regard to digital media, we need this focus on a new materialism. In the hidden abode of contemporary production, the medium/matter distinction hasn't disappeared, but has simply been reorganized. We could use the distinction between the medium of storage and the medium of display as an illustration. Within analog culture, there is usually a causal relationship between storage and display, say between negative and print, but in algorithmic culture, the relationship has become not simply arbitrary, but dependent on the new interstice of software, or what I would call an algorithmically enabled work process. By using the term "algorithmic culture", I want to direct attention to this new and fundamental

difference. An algorithmic culture is not an immaterial culture, but a factory where the work process abides in different layers of code and algorithms.

If we juxtapose the two images, the old Niepce heliograph and Müller-Pohle's *Digital Scores*, the question is, what controls the move or the transition between these two images? By using information theory, we can say that we see two types of information and two different types of media. They both direct attention toward a material substratum of production: dots in the case of Niepce and numbers in the case of Müller-Pohle. We move from dots to digits. If we formalize the impression we can put it this way: it is the same image, but represented differently. The first we would call a photograph, the second is a numerical representation of the photograph. The identity of media itself has changed fundamentally. Put more simply, we could say that in the case of Niepce, there is a causal relationship between storage and display, but in Müller-Pohle's case, the new storage medium of digital codes has itself become what is displayed. Müller-Pohle's experiment addresses the new "non-visual" layer of visual culture by arresting the technological nature of image processing at the stage "before" it becomes an image in the ordinary sense. He implicitly criticizes Bolter and Grusin's idea that the digital image essentially remediates earlier media, for example, realistic photography or analog film, but the heart of the matter is more complex: the diffusion of informatics within the production and recirculation of images actually – and essentially – negates and postpones the phenomenological output "level" of the image. In a paradoxical maneuver, Müller-Pohle displays the non-visual machinery of the before-the-image as a multiple, as eight potential images, as codes to be reprogrammed. The machinic ground of the image (codes and algorithms) has become a reservoir for a plurality of expressions.²⁰ The digital image becomes a new (non)ground for unprecedented spatio-temporal explorations.

The Image as Projectile

Ken Jacobs has used both analog and digital technologies in his found-footage film experiments. His *TOM, TOM, THE PIPER'S SON* (1969/1970) rephotographed Billy Bitzer's ten-minute silent short from 1905 into a 110-minute delayed exploration of the material. He uses what could be called an analog technique (rephotographing off the screen) to create a vertical effect. A recurring stylistic effect is namely the ways in which he digs deeply into each frame of the film to excavate the hidden life of the grain of the image.²¹ Jacobs' film is widely recognized as a classic structural film and was added to the US National Film Registry by the Library of Congress in 2007. In 2008, Jacobs reworked Bitzer's old

short with the aid of digital tools. He calls this version, *RETURN TO THE SCENE OF THE CRIME*. This time he limits it to reworking the film's short opening scene where the pig is stolen. However, this version is 93 minutes long, but instead of digging deeply into the frames in a vertical manner, he uses a technique that is comparative and horizontal. It is not primarily the grain of the image that he studies here; his comparative analysis of images and sequences often juxtaposes images or a series of images in different technical (digital) attires. In a fabulous sequence, the juggler's balls are transformed into still images "captured" from different fractions of the opening scene (fig. 1). It is as if the whole opening scene is folded into a spectacular explosion of past and present moments, creating an intensive time-space. The juggler scene suddenly comes to a halt, is then run backwards and then forward again. The sequence intensifies time and space in ways that eclipse the analog version of the same material.



Fig. 1. Still from Ken Jacobs's *RETURN TO THE SCENE OF THE CRIME* (2008)

These intensities have been further explored by Jacobs' student Gregg Biermann, who also refashions found footage, often using Hitchcock classics. I call his works "software cinema" because they so insistently explore a found sequence of film images according to a preprogrammed software feature or "spe-



Fig. 2. Still from Gregg Biermann's LABYRINTHINE (2010)

cial effect". In LABYRINTHINE (2010) Biermann works with 41 memorable and iconographic shots in Hitchcock's classic VERTIGO (1958). The shots are superimposed on top of each other, creating a hypnotic labyrinth of repetitions and transformations (fig. 2). The different moments overlap in a kind of contrapunctual proliferation lasting 15 minutes. This superimposition of images is unlike how we remember it from the classical (analog) techniques, where the images are partly transparent. The images are composite sequences of concentric rectangles. The rectangular screen no longer frames one shot at a time, rather, the screen becomes a theatre for a multiplicity of images where each new shot is born within the previous shot as a new rectangle, which gradually increase in size and finally covers the last shot. As it grows, new shots are born within the shot and this goes on according to a rhythmical scheme where each shot is repeated four or five times. As the film develops, several series of shots overlap. Cinematic motion as the movement of objects in space *within* the image is here competing with the movement *between* blocks of floating images. The blocks float like moving pictures *through* the screen like an approaching bullet or projectile. Ultimately, a labyrinth of movements appears both within the image (the image within the image) and between the images (the changing relationship between the images within the image). Continuity editing is replaced by a discontinuous and labyrinthine editing process, and the screen no longer displays one image at a time, but several.

In spite of the rhythmical slowness of the floating concentric rectangles, it is sometimes confusing to distinguish between them as if this virtual multiplicity

of time folding images *indexes* in intangible ways the constant confusion and dizziness of the protagonist of the film, Scottie (James Stewart) – and the viewer – who both are sometimes unable to distinguish between the different women (the woman remembered and the woman seen) as played by Kim Novak. Biermann's hypnotic repetition and manipulation of the characteristic soundtrack together with the "floating" iconography of the film highlights the entire repertoire of genres at play in *VERTIGO*, which itself floats in between a detective mystery thriller, a romantic melodrama and a horror movie. Most of the shots are of Scottie as he is observing or thinking; he hardly moves during these sequences. The images have a stuttering and discontinuous logic that arrests and focuses on perceptual phenomena, which slips away when we view the film. Biermann opens these intervals of the imperceptible, enlarges them and turns them into an art form of their own. It is as if he is using the technologically enhanced quality of the images to explore the optical unconscious in a way never envisioned by Walter Benjamin.



Fig. 3. Still from Gregg Biermann's *SPHERICAL COORDINATES* (2005)

This feature of the unconscious is taken to a different vein in *SPHERICAL COORDINATES* (2005), which remakes a short scene from Hitchcock's *PSYCHO*. Here, we concentrate on Janet Leigh who is driving. The image twists, bends, folds and turns around (sometimes still, sometimes moving) as she steers the wheel (fig. 3). "The camera moves in a variety of ways, examining the inside of a 3D

animated sphere on the inside of which a scene from *PSYCHO* is wrapped”, explains Biermann.²² It is as if a camera is analyzing the manipulation of the image itself. *SPHERICAL COORDINATES* and *LABYRINTHINE* foreground the digital imaging tools in ways that turn the films into sensitive systems. He manipulates well-known found footage, that is, he intervenes in our memory, a collective system where images have become “emoticons” of sorts, and alters the very threads of this system. The exploration of abnormal movements between the still and the moving image create a system of post-cinematic “emoticons” mapping and altering the coordinates of a sensitive system which is collective or more than human.

The Time of Technology

These new spatio-temporal explorations seem to reveal a new potential for the visual. We need to question the relationship between images and the body, between the inhuman and the human in new ways. I will show how Mark B.N. Hansen does this with regards to the digital image, but first, Rodowick’s critique of the digital image. He laments the inability of the digital image to convey an experience of time passing, and he continues:

It might be clarifying at this point to insist on the rather radical point that in the digital universe there is no cinema, no photography, no images, and no sounds. The cosmogony of computers only recognizes symbolic notation and algorithmic operations, and is totally agnostic as to outputs. We are used to thinking of images on the model of painting or photography, as limited extensions of space present to us as a whole, and we want to think of digital or electronic “images” in this familiar way. But an electronic image is not “one” – it is never wholly present to us because screens are being constantly refreshed and rewritten. And again, the code writing this output to the screen can just as easily take the form of text, a sound, or an abstract symbolic notation. This is why I say in *The Virtual Life of Film* that the reconstitution of an image from digital information is something like making a very detailed painting from the information given in a very precise description. Or, to refer to a wonderful early text by Roland Barthes, digital “photography” cannot be considered as a “message without a code” – it is only code and nothing else.²³

Rodowick poses a radical opposition between the analog and the digital image. He asserts that the radical otherness of the digital image creates a “naming crisis”. We need a new name for the image, or for the no-longer-an-image-as-we-used-to-know-it. In *The Virtual Life of Film*, Rodowick argues that the specific quality of the analog image resides in the fact that it conveys a unique impres-

sion of time passing due to the fact that there is a causal relationship between input (the time of shooting) and output (projection). The digital image essentially lacks this causal relationship. Its multiplicity resides in its discontinuity between input and output. Furthermore, Rodowick establishes a causal relationship between the technical quality of the image and a certain mode of experience. One image conveys a certain experience of time, another does not. Significantly, Hansen establishes a similar causal relationship between technology and experience, but he reaches the opposite conclusion.

As I see it, digitization requires us to reconceive the correlation between the user's body and the image in an even more profound manner. It is not simply that the image provides a tool for the user to control the "infoscape" of contemporary material culture, as Manovich [and Rodowick] suggests, but rather that the "image" has itself become a process, and, as such, has become irreducibly bound up with the body.²⁴

Hansen's new embodied understanding of the digital image relies on a concept of affectivity, which he describes as "the capacity of the body to experience itself as 'more than itself' and thus to deploy its sensory-motor power to create the unpredictable, the experimental, the new" (7). Both Hansen and Rodowick use a modern phenomenology to posit their argument, but their phenomenology seems to be based on two different conceptions of technology and the self. To Rodowick, the analog image becomes a time traveler negotiating between two experiences of time, but the digital image is unable to capture this experience of duration because it is never "wholly present to us"; it is merely a contingent configuration of numerical values where an indexical basis is replaced by a continual scanning or updating, he would assert. Hansen, on the other hand, seeks to implicate the process of image making itself as a certain time of technology where information is "made perceivable through embodied experience" in new ways.²⁵

Upon viewing Biermann's *LABYRINTHINE*, both Hansen and Rodowick offer significant insights into the analysis of the experience. On the one hand, Hitchcock's *VERTIGO* is replaced by a new "feel" of the image. The film no longer carries the same kind of aura as we experienced in the original theatrical release. However, the ground of this aura (its originary memory impact, so to speak) is addressed as a malleable nervous system (both emotional and visual). Rather than a photochemical "index", the experience of computer-generated image processing opens up the potential for a new experience of the film as a way of *indexing* the changing vocation of the life of the new media image itself. This new opening relates less (or not at all, as Rodowick would perhaps contend) to the kind of time or duration at play in an analog image, but it foregrounds a new technicity of the image which exposes time as a non-human nervous system or a becoming. In other words, the loss of a certain "feeling" of "human"

time (Rodowick) may be the necessary condition for the new image to produce the mode of affective experience addressed by Hansen.

Kaleidoscope

The multiplicity of the digital image can be studied even more dramatically in the work by new media artists such as Mario “Quasimodo” Klingemann. He experiments with new algorithms and software on his website, which can be downloaded or simply interacted with. Some, like his piece *Kaleidoscope*, are pieces of software that can accept a painting or a still photograph as an input and a moving or an interactive image as an output. As soon as one engages with *Kaleidoscope*, the website takes complete control of the user’s mouse and keyboard. Each key instantiates an algorithm, which alters the image. The old film/photo divide is deconstructed and integrated into a playful structure of interactivity. The viewer becomes a player interacting or infecting the image itself with movement. When the Web site is opened, *Kaleidoscope* presents a stilled wheel with images of a woman. The image is based on J.H. Lynch’s drawing “Autumn Leaves”. Once you click on the image, the keyboard is activated and “she” begins to move according to the following parameters:

- [1] – start/stop rotation 1
- [2] – start/stop rotation 2
- [3] – start/stop rotation 3
- [4] – toggle flip tiles
- [+]/[-] – more/less tiles
- [q]/[w] – rotation 1 speed
- [a]/[s] – rotation 2 speed
- [y]/[x] – rotation 3 speed
- [e] – reset rotation 1 speed
- [d] – reset rotation 2 speed
- [c] – reset rotation 3 speed
- [UP]/[DOWN] – zoom in/out
- [m]/[n] – spiral offset
- [o] – reset spiral offset
- [5]/[6] – shearing 1
- [7]/[8] – shearing 2.²⁶

The process of interacting with the image is inseparable from what Hansen calls affectivity. This is an experimental becoming where the body of the player deploys “its sensory-motor power to create the unpredictable, the experimental,

the new".²⁷ We don't view the image, we play with it, and each action activates an algorithm which instantiates a certain move and a certain speed of the image, as well as simulated camera moves. However, the algorithms also reveal themselves in the transitions between arrest and motion and in between different motions, etc. This kind of interaction with what looks like a still image is a feature of many of Klingemann's web installations, such as several of his so-called *Incubator* sketches, like *Rubber Screen*.

Klingemann's generative art work *Flickeur* is a software-based "machine" which intervenes or interacts with the database Flickr.com. Once the user/viewer presses play, *Flickeur* will randomly begin to retrieve images from Flickr.com and create an infinite "film" with a style that can vary between stream-of-consciousness, documentary or video clip. All the blends, motions, zooms or time leaps are completely random. *Flickeur* works, according to Klingemann, "like a looped magnetic tape where incoming images from Flickr.com will merge with older materials and be influenced by the older recordings' magnetic memory. The virtual tape will also play and record forward and backward to create another layer of randomness."²⁸ Klingemann actually programs a series of random errors and glitches to simulate the appearance of the wear and tear of a filmstrip or a magnetic band being over-written.²⁹ These marks and the nerve-racking soundtrack sometimes turn the material into a very suggestive horror story. Here, www.flickr.com, which is one of the largest user-based digital photo archives in the world, is transformed into a thriller.

One could argue that the examples in this article are not representative of what we call the digital image. I think we need to go beyond many of the most popular theories about the digital image, such as Bolter's and Grusin's notion that new media basically remediates older media.³⁰ This is, as I have shown, insufficient. Rodowick, Hansen and Manovich's approaches (despite their internal disagreements) seem more promising. If we take into account that the characteristics of the digital image essentially reside in its processual aspects rather than in its product aspects, the examples in this chapter are exemplary. New media tools can be used to analyze old media in the fashion preferred by Laura Mulvey, but ultimately the digital image as envisioned by Hansen and Rodowick, and demonstrated by Müller-Pohle, Jacobs, Biermann and Klingemann, speak of a different image. It is connected to a new technical matrix, an algorithmic culture, which interrupts the image not to analyze it in its unique singularity, but to give birth to a potential multiplicity which is always more than one – and ready to change.

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Notes

1. See D.N. Rodowick, *The Virtual Life of Film* (Cambridge, MA: Harvard University Press, 2007).
2. Stanley Cavell, *The World Viewed* enlarged ed. (Cambridge, MA: Harvard University Press, 1979), ix.
3. Laura Mulvey, *Death 24x a Second: Stillness and the Moving Image* (London: Reaktion Books, 2006), 7.
4. *Ibid.*, 146.
5. See, for example, the works of Henry Jenkins, Thomas Elsaesser and David Bordwell, who basically agree on the fact that Hollywood's way of narrating stays the same, but more significantly, they agree that Hollywood can narrate films with greater complexity and oddness due to computer-generated imaging and further, they can expand their narratives through what Jenkins calls the business of "trans-media". See Jenkins, *Convergence Culture: Where Old and New Media Collide* (New York: New York University Press, 2006) and Thomas Elsaesser, "The Digital Paradox", unpublished lecture, 2010.
6. Manovich, *The Language of New Media* (Cambridge, MA: MIT Press, 2001).
7. As Les Goldschlager and Andrew Lister write in their classic textbook on computer science: The algorithm "is the unifying concept for all the activities which computer scientists engage in" (Goldschlager and Lister, *Computer Science: A Modern Introduction* (New York: Prentice Hall, 1988), 1.
8. Manovich, *Software Takes Command*. Available at <http://www.manovich.net/> (accessed July 12, 2010).
9. D.N. Rodowick, *The Virtual Life of Film* (Cambridge, MA: Harvard University Press, 2007) 110.
10. *Ibid.*, 130.
11. William Uricchio: "The Algorithmic Turn: Photosynth, Augmented Reality and the Changing Implications of the Image" (unpublished manuscript).
12. See Alexander Galloway, *Gaming: Essays on Algorithmic Culture* (Minneapolis: University of Minnesota Press, 2006).
13. Galloway, *Gaming*, 6.
14. *Ibid.*, 14.
15. Alexander Galloway. "What You Get Is What You See" in *The Archive in Motion. New Conceptions of the Archive in Contemporary Thought and New Media Practices*. Nota bene: Studies from the National Library of Norway, E. Røssaak (ed.) (Oslo: Novus, 2010).
16. Geoffrey Batchen, *Each Wild Idea: Writing, Photography, History* (Cambridge, MA: MIT Press, 2002) 177.
17. Margaret Dikovitskaya, "An interview with Laura U. Marks". In Margaret Dikovitskaya, *Visual Culture: A Study of the Visual After the Cultural Turn*. (Cambridge, MA: MIT Press, 2005), p. 215-216.
18. Batchen, *Each Wild Idea*, 179.
19. Friedrich Kittler, "Computeranalphabetismus" in F. Kittler, *Short Cuts* (Frankfurt am Main: Zweitausendeins, 2002), 109-133.

20. I play here upon Jean-Luc Nancy's *The Ground of the Image* (New York: Fordham University Press, 2005). He talks about the ground (*fond*) of the image as that which "appears as what it is by disappearing" (p. 7). In the algorithmic culture, the ground of the image must be rethought, because the ground no longer "passes entirely into the image" as Nancy asserts. An important philosophical inspiration for me has been Gille Deleuze's idea about multiplicity as a complex structure that does not refer to a prior unity, but rather *is* a becoming. The digital image and even more so, the generative art image, is exemplary as a "sensory becoming": "sensory becoming is otherness caught in a matter of expression" (Deleuze and Guattari, *What is Philosophy?* (London: Verso, 1994), 177).
21. This is analyzed more fully in E. Røssaak: *The Still/Moving Image: Cinema and the Arts* (Saarbrücken: Lambert Academic Press, 2010).
22. Biermann quoted from his website, <http://www.greggbiermann.com/spherical.htm> (last accessed September 10, 2010).
23. Rodowick. "Thinking Virtual", lecture at University of Chicago, April 6, 2010.
24. Hansen, *New Philosophy for New Media* (Cambridge, MA: MIT Press) 10.
25. Ibid.
26. Listed on the website: <http://www.quasimondo.com/archives/000579.php> (last accessed October 10, 2010).
27. Hansen, p. 7.
28. Klingemann on <http://www.quasimondo.com/archives/000579.php> (last accessed October 10, 2010).
29. "I use the term 'Magnetic' more as a Metaphor to describe what a theoretical machine would work like if I'd be able to rebuild the software engine behind flickeur as a physical real-world model.... The equivalent of the 'magnetic tape' inside my program is just an area in the computer's memory (RAM). And since computers are unfortunately mostly perfect when it comes to reading and writing to memory I have to introduce the glitches and errors myself that happen when old data is merged with new data (image data in this case). For that purpose I have written a library of bitmap manipulation algorithms that try to create a big variety of visual effects. What was very important for me here is that I want the computer to be able to surprise me with unexpected visual output. Also there is the issue that computers cannot generate true randomness by themselves – there are random number generators but those are still all based on mathematical formulas which will create the same sequence of numbers for a given seed number. So the true randomness in this case comes from the outside in the form of the images. An image in a computer is just a big list of numbers and even though in a normal picture those numbers have a certain order they still can be considered as 'random' since looking at just a single pixel or number in an image it would be impossible to say what the next number will be." Klingemann in an email to the author, October 22, 2010.
30. Bolter and Grusin, *Remediation: Understanding New Media* (Cambridge, MA: MIT Press, 1999).

